Ahmed Nasser

Data Engineer & Data Analyst

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EDUCATION

Cairo University, Faculty of Computers and Artificial Intelligence Expected 2029
B.Sc. in Computer Science – Information Systems (IS) Major
Currently a 2nd-year student, GPA: 3.1/4.0, Very Good

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, HTML5 / CSS3
Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, Excel, EDA, Data Cleaning, Statistical Analysis
Data Engineering: ETL / ELT Pipelines, Data Modeling, Data Warehousing, Star Schema, SQLAlchemy, Apache Airflow, n8n
Big Data: Hadoop, HDFS, MapReduce, Apache Spark, PySpark, Hive, Kafka, Batch & Stream Processing
Databases & Tools: PostgreSQL, Docker, Git & GitHub, VS Code, Jupyter Notebooks, Linux (Kali / Ubuntu)

WORK EXPERIENCE

Big Data Intern September 2026 – Present
Huawei Ministry of Youth Academy *Cairo, Egypt*
- Learning the foundations of Big Data (the 5 Vs, distributed storage, parallel processing) and the Hadoop ecosystem (HDFS, MapReduce, YARN) for large-scale datasets.
- Practicing large-scale processing with Apache Spark and Hive, building SQL and PySpark workflows to clean, transform, and query structured and semi-structured data.
- Exploring batch and real-time ingestion pipelines and data warehouse architectures, applying best practices for scalable, reliable pipelines that feed analytics.

Data Analyst Trainee 2026 – Present
Digital Egypt Pioneers Initiative (DEPI) *Cairo, Egypt*
- Extracting, cleaning, and preparing multi-source datasets for exploratory data analysis and actionable insights.
- Building interactive data visualization dashboards and automated reporting pipelines.
- Applying data analytics frameworks, statistical techniques, and ETL processes using Python and SQL.

PROJECTS

Customer 360 Analytics *Python, Pandas, NumPy, Seaborn, Matplotlib*
- Cleaned and merged 9 raw CSV files into a single Customer 360 table of 1,500 customers and 31 features, resolving mixed date formats, currency strings, duplicate records, and city-country mismatches.
- Engineered KPIs (65.1% active rate, 24.1% churn rate, ~$990 ARPU) and answered 19 business questions with Seaborn and Matplotlib visuals, ending in data-backed recommendations on retention, support quality, and plan upselling.

Superstore Data Warehouse *PostgreSQL, SQL, Star Schema, ETL*
- Built a PostgreSQL warehouse that loads raw retail data into a staging layer, then a star schema (1 fact + 4 dimension tables) with primary/foreign keys, join and date indexes, a denormalized reporting view, and a PL/pgSQL stored procedure for yearly KPIs.
- Wrote 16 analytical queries using CTEs and window functions (LAG, RANK, NTILE, running totals) covering profitability, discount impact, RFM scoring, month-over-month growth, and return rates; verified performance with EXPLAIN ANALYZE.

Baby Photoshop – C++ Image Processor *C++, Qt, Variadic Templates*
- Developed a desktop image-processing application with 18 filters implemented from scratch on raw pixel data, including geometric transforms, tonal adjustments, blur, edge detection, and creative effects.
- Designed a reusable variadic-template pipeline to load, filter, and save images, keeping each filter independent and easy to extend, and added a Qt GUI.

Medical Identical Trace (Med Trace) *System Design, Database Design, Data Analytics*
- Designed a national digital medical-record platform where citizens register with their national ID through a mobile app, and diseases, doctor visits, and surgeries are recorded across their lifetime to support diagnosis, emergency access, and national health statistics.

Monster Wrangler Game *Python, Pygame, OOP*
- Built a 2D arcade game with object-oriented game state management, sprite animation, dynamic collision handling, and progressive difficulty.

Interstellar – Cinematic Movie Website *HTML5, CSS3, JavaScript*
- Built a multi-page front-end site with a custom vinyl-record soundtrack player (HTML5 Audio API, vanilla JavaScript), a split-panel Login/Register page, and a Crew page with glassmorphism cast cards.

CERTIFICATIONS

- IBM Data Fundamentals – IBM SkillsBuild
- IBM Data Science Professional Certificate – IBM / Coursera
- CS50's Introduction to Databases with SQL – Harvard University